

2520030340

EXPERIMENT – 7

Title:

Process Synchronization and Child Process Monitoring Using `waitpid()`

Aim:

To implement process synchronization using `waitpid()`, monitor child processes, retrieve the `PATH` variable, locate executable commands, verify permissions, and handle missing commands.

Objective:

- To understand process synchronization.
- To monitor child processes.
- To use `waitpid()` for synchronization.
- To retrieve the `PATH` environment variable.
- To search directories in `PATH`.
- To locate executable files.
- To verify file permissions.
- To handle missing commands.
- To test command resolution.

Theory:

`waitpid()` is used by a parent process to wait for a specific child process to finish.

The `PATH` environment variable contains a list of directories where executable programs are located.

For example, when the user enters:

ls

the shell searches the directories in `PATH` to find the executable.

The shell checks whether the command exists and whether it has execute permission. If the command is not found, an error message is displayed.

Algorithm:

1. Start the shell.
2. Retrieve the `PATH` environment variable.
3. Read the command entered by the user.
4. Parse the directories stored in `PATH`.
5. Search each directory for the command.
6. Check whether the executable exists.
7. Check whether the file has execute permission.
8. Create a child process using `fork()`.
9. Execute the command using `execvp()`.
10. Use `waitpid()` in the parent process to wait for the child.
11. Handle missing commands and execution errors.
12. Repeat until the user enters `exit`.
13. Stop.

Functions Used:

Function	Purpose
<code>getenv()</code>	Retrieve <code>PATH</code>
<code>strtok()</code>	Parse directories
<code>access()</code>	Check file and execute permissions
<code>fork()</code>	Create child process
<code>execvp()</code>	Execute command
<code>waitpid()</code>	Wait for a specific child
<code>printf()</code>	Display output

```
perror(    Display errors
)
```

Sample Input:

```
2520030340_kruthi> echo hello
2520030340_kruthi> pwd
2520030340_kruthi> ls
2520030340_kruthi> xyzabc
```

Sample Output:

```
hello
/home/kruthi
...
Command not found
```

Observation:

1. The `PATH` variable was retrieved successfully.
2. Commands were searched in the `PATH` directories.
3. Executable files were located successfully.
4. File permissions were checked.
5. Child processes were created successfully.
6. The parent process waited for the child using `waitpid()`.
7. Missing commands were handled correctly.

Result:

The shell successfully performed **PATH-based command resolution and child process synchronization** using `waitpid()`.

Conclusion:

The experiment successfully demonstrated **child process monitoring and synchronization** using `waitpid()`. The shell also retrieved the `PATH` variable, searched directories, checked executable permissions, and handled missing commands.

```
kruthi@LAPTOP-LP9QU3RR: ~  
kruthi@LAPTOP-LP9QU3RR:~$ pwd  
/home/kruthi  
kruthi@LAPTOP-LP9QU3RR:~$ ls  
OSSP                                program.c  
Prime-osp-os                        shell  
directory_name                      shell.c  
os-operating-system-and-system-programming-OSSP student.txt  
process.c.save                     text.txt  
program  
kruthi@LAPTOP-LP9QU3RR:~$ date  
Tue Sep 22 05:23:30 UTC 2026  
kruthi@LAPTOP-LP9QU3RR:~$ xyzabc  
xyzabc: command not found  
kruthi@LAPTOP-LP9QU3RR:~$ echo hello world  
hello world  
kruthi@LAPTOP-LP9QU3RR:~$ ./shell  
2520030340_kruthi> echo $PATH  
$PATH  
Child process 518 exited with status 0  
2520030340_kruthi> pwd  
/home/kruthi  
Child process 519 exited with status 0  
2520030340_kruthi> ls  
OSSP                                program.c  
Prime-osp-os                        shell  
directory_name                      shell.c  
os-operating-system-and-system-programming-OSSP student.txt  
process.c.save                     text.txt  
program  
Child process 520 exited with status 0  
2520030340_kruthi> xyzabc  
Command not found: xyzabc  
2520030340_kruthi> date  
Tue Sep 22 05:24:27 UTC 2026  
Child process 521 exited with status 0  
2520030340_kruthi> ./shell  
2520030340_kruthi> pwd  
/home/kruthi  
Child process 527 exited with status 0  
2520030340_kruthi>  
2520030340_kruthi> ls  
OSSP                                os-operating-system-and-system-programming-OSSP program.c student.txt  
Prime-osp-os                        process.c.save shell text.txt  
directory_name                      program shell.c  
Child process 528 exited with status 0  
2520030340_kruthi> date  
Tue Sep 22 05:25:14 UTC 2026  
Child process 529 exited with status 0
```



kruthi@LAPTOP-LP9QU3RR: ~



kruthi@LAPTOP-LP9QU3RR:~\$ pwd

/home/kruthi

kruthi@LAPTOP-LP9QU3RR:~\$ ls

OSSP

Prime-oss-p-os

directory_name

os-operating-system-and-system-programming-OSSP

process.c.save

program

kruthi@LAPTOP-LP9QU3RR:~\$ date

Tue Sep 22 05:23:30 UTC 2026

kruthi@LAPTOP-LP9QU3RR:~\$ xyzabc

xyzabc: command not found

kruthi@LAPTOP-LP9QU3RR:~\$ echo hello world

hello world

kruthi@LAPTOP-LP9QU3RR:~\$./shell

2520030340_kruthi> echo \$PATH

\$PATH

Child process 518 exited with status 0

2520030340_kruthi> pwd

/home/kruthi

Child process 519 exited with status 0

2520030340_kruthi> ls

OSSP

Prime-oss-p-os

directory_name

os-operating-system-and-system-programming-OSSP

process.c.save

program

Child process 520 exited with status 0

2520030340_kruthi> xyzabc

Command not found: xyzabc

2520030340_kruthi> date

Tue Sep 22 05:24:27 UTC 2026

Child process 521 exited with status 0

2520030340_kruthi> ./shell

2520030340_kruthi> |

program.c

shell

shell.c

student.txt

text.txt

program.c

shell

shell.c

student.txt

text.txt



kruthi@LAPTOP-LP9QU3RR: ~



```
kruthi@LAPTOP-LP9QU3RR:~$ pwd
/home/kruthi
kruthi@LAPTOP-LP9QU3RR:~$ ls
OSSP                                program.c
Prime-osp-os                        shell
directory_name                     shell.c
os-operating-system-and-system-programming-OSSP student.txt
process.c.save                     text.txt
program
kruthi@LAPTOP-LP9QU3RR:~$ date
Tue Sep 22 05:23:30 UTC 2026
kruthi@LAPTOP-LP9QU3RR:~$ xyzabc
xyzabc: command not found
kruthi@LAPTOP-LP9QU3RR:~$ echo hello world
hello world
kruthi@LAPTOP-LP9QU3RR:~$ ./shell
2520030340_kruthi> echo $PATH
$PATH
Child process 518 exited with status 0
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2520030340_kruthi> ./shell
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Prime-osp-os                        process.c.save shell text.txt
directory_name                     program shell.c
Child process 528 exited with status 0
2520030340_kruthi> date
Tue Sep 22 05:25:14 UTC 2026
Child process 529 exited with status 0
2520030340_kruthi> xyzabc
Command not found: xyzabc
2520030340_kruthi>
```